Asymptotics and consistency

EC 421, Set 6

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Prologue

Schedule

Last Time

Living with heteroskedasticity

Today

Asymptotics and consistency

Next

- New problem set
- *Next topic:* Time series
- Midterm next week (May 7th)

R showcase

Need speed? R allows essentially infinite parallelization.

Three popular packages:

- `future` and `furrr`
- `parallel`
- `foreach`

And here's a nice tutorial.

Consistency

Consistency

Welcome to asymptopia

Previously: We examined estimators (*e.g.*, $\hat{\beta}_j$) and their properties using

Consistency

Welcome to asymptopia

New question:

How does our estimator behave as our sample gets larger (as $n \to \infty$)?

Consistency

Probability limits

Just as the *expected value* helped us characterize **the finite-sample distribution of an estimator** with sample size n,

1. Here, "big" n means $n \to \infty$. That's *really* big data.

Consistency

Probability limits

Let B_n be our estimator with sample size n.

Then the **probability limit** of B is α if

$$\lim_{n \to \infty} P\left(|B_n - \alpha| > \epsilon\right) = 0 \tag{1}$$

for any $\varepsilon > 0$.

Consistency

Probability limits

Equivalent statements:

- The probability limit of B_n is α.
- $\text{plim}\, B = \alpha$
- B converges in probability to α.

Consistency

Probability limits

Probability limits have some nice/important properties:

- $\text{plim}\,(X \times Y) = \text{plim}\,(X) \times \text{plim}\,(Y)$
- $\text{plim}\,(X + Y) = \text{plim}\,(X) + \text{plim}\,(Y)$
- $\text{plim}\,(c) = c$, where c is a constant
- $\text{plim}\left(\dfrac{X}{Y}\right) = \dfrac{\text{plim}\,(X)}{\text{plim}\,(Y)}$
- $\text{plim}\big(f(X)\big) = f\big(\text{plim}\,(X)\big)$

Consistency

Consistent estimators

We say that **an estimator is consistent** if

1. The estimator **has a prob. limit** (its distribution collapses to a spike).
2. This spike is **located at the parameter** the estimator predicts.

Consistency

Consistent estimators

Example: We want to estimate the population mean μ_x (where X~Normal).

Let's compare the asymptotic distributions of two competing estimators:

1. The first observation: X_1
2. The sample mean: $\overline{X} = \frac{1}{n} \sum_{i=1}^{n} x_i$
3. Some other estimator: $\widetilde{X} = \frac{1}{n+1} \sum_{i=1}^{n} x_i$

Note that (1) and (2) are unbiased, but (3) is biased.

Consistency

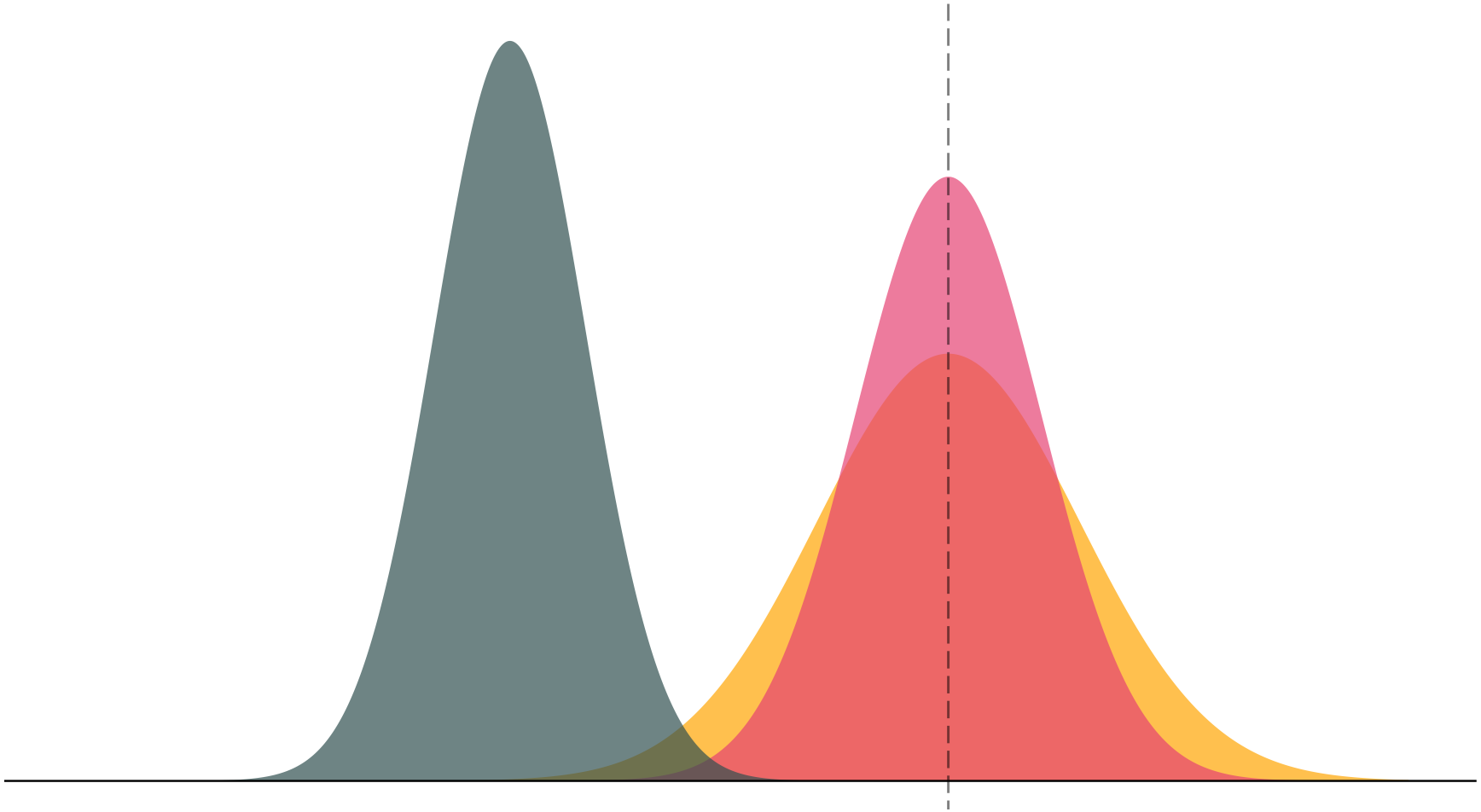
Consistent estimators

To see which are unbiased/biased:

Consistency

Distributions of X_1, $\overline{X}$, and $\widetilde{X}$

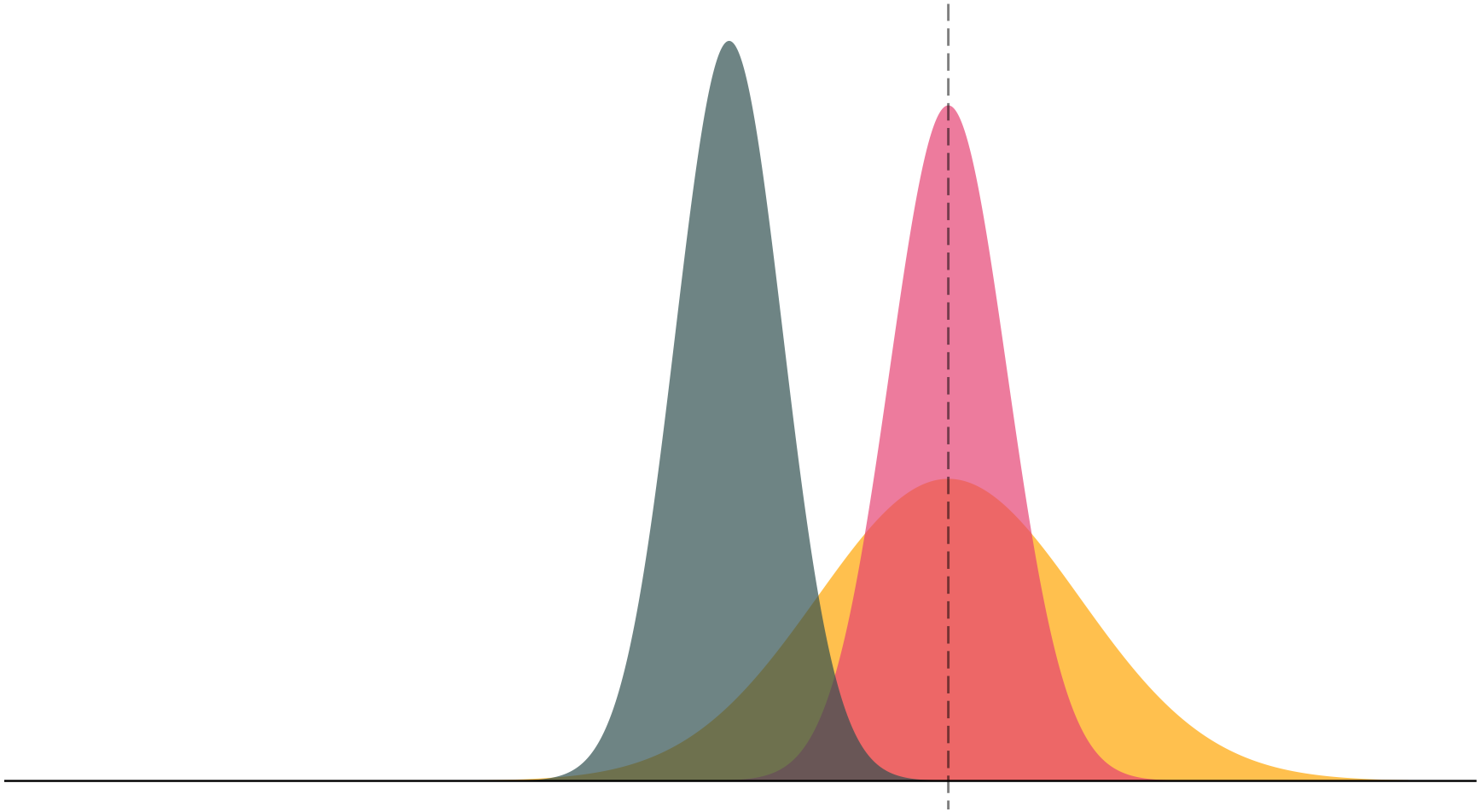
$n = 2$



Consistency

Distributions of X_1, $\overline{X}$, and $\widetilde{X}$

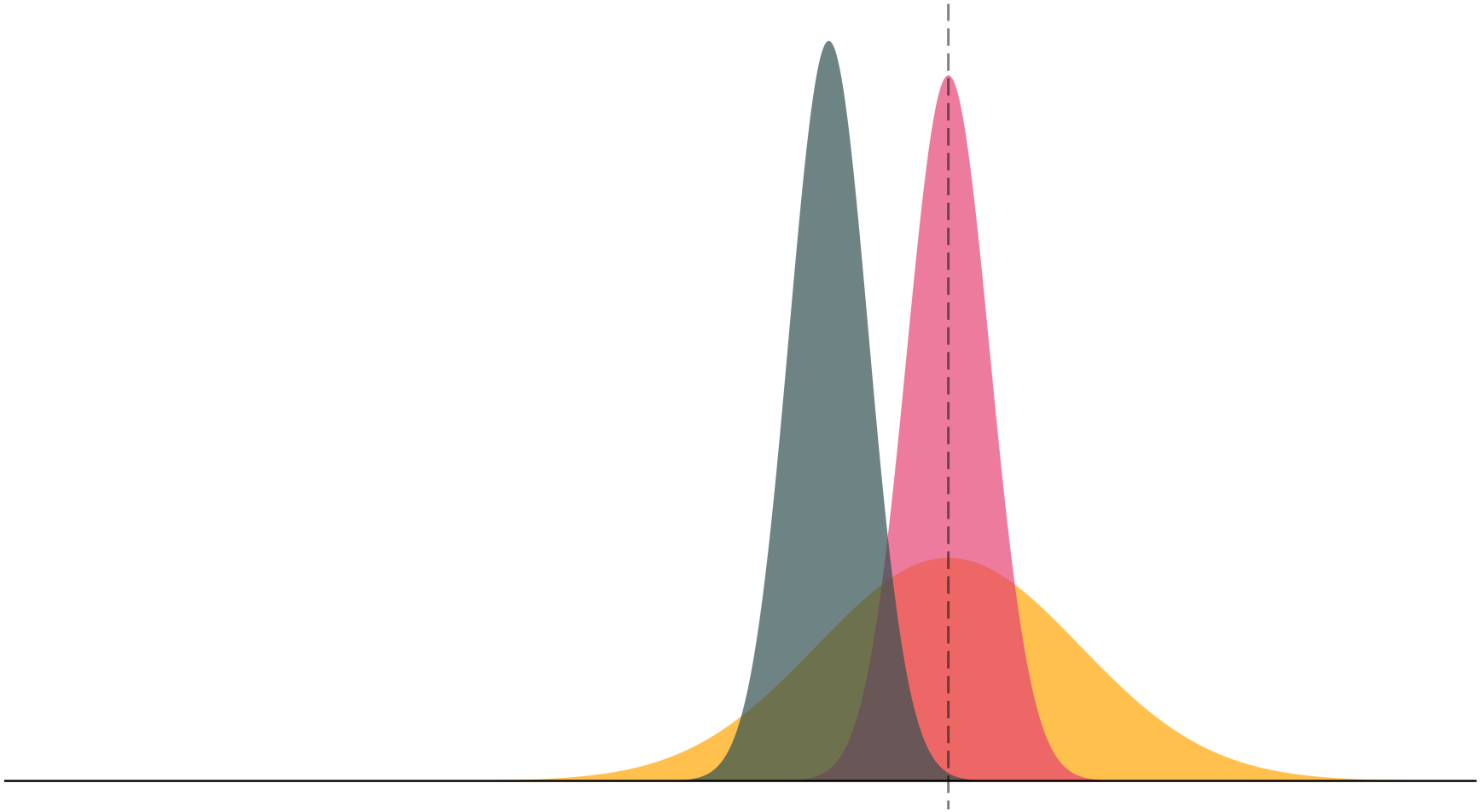
$n = 5$



Consistency

Distributions of X_1, $\overline{X}$, and $\widetilde{X}$

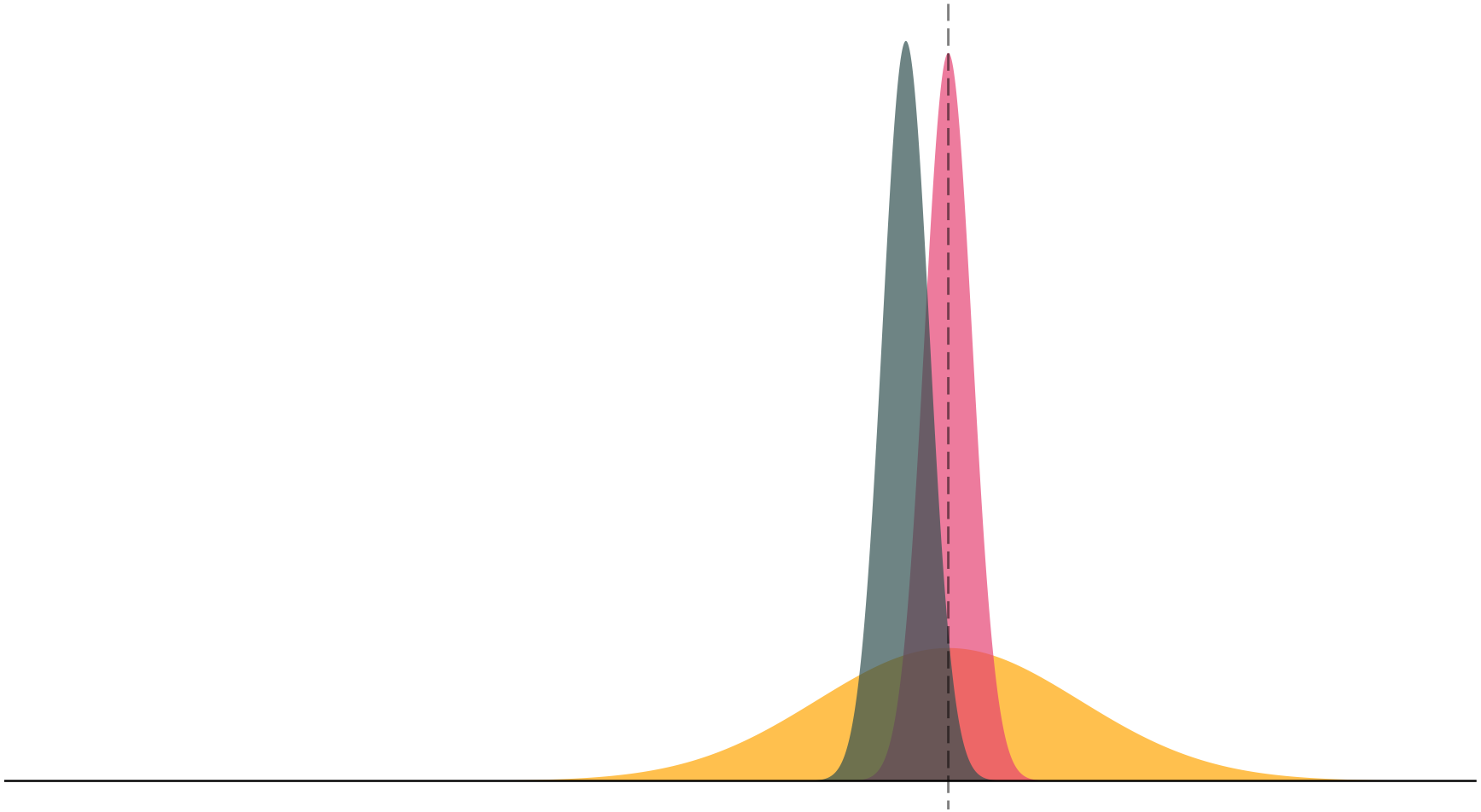
$n = 10$



Consistency

Distributions of X_1, $\overline{X}$, and $\widetilde{X}$

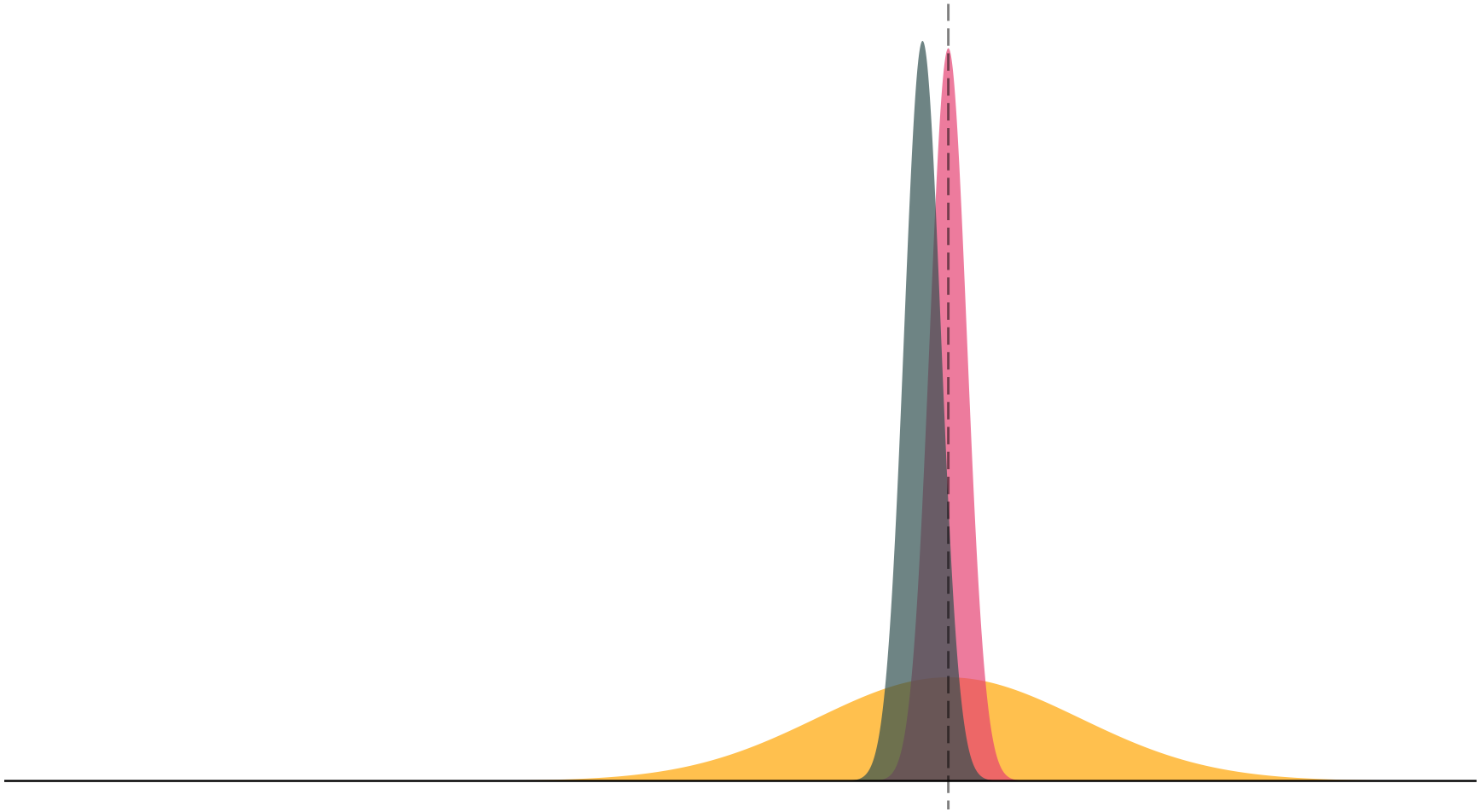
$n = 30$



Consistency

Distributions of X_1, $\overline{X}$, and $\widetilde{X}$

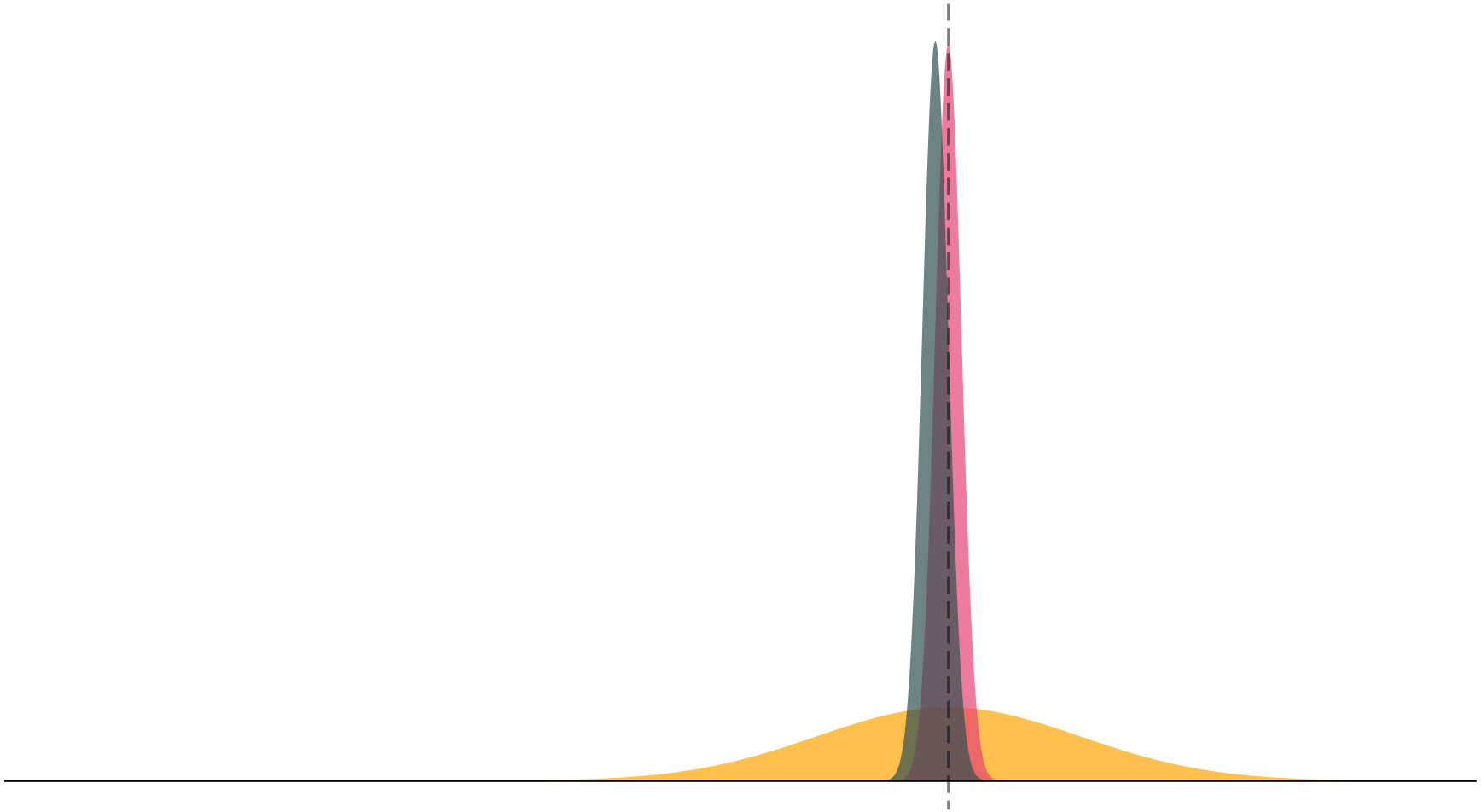
$n = 50$



Consistency

Distributions of X_1, $\overline{X}$, and $\widetilde{X}$

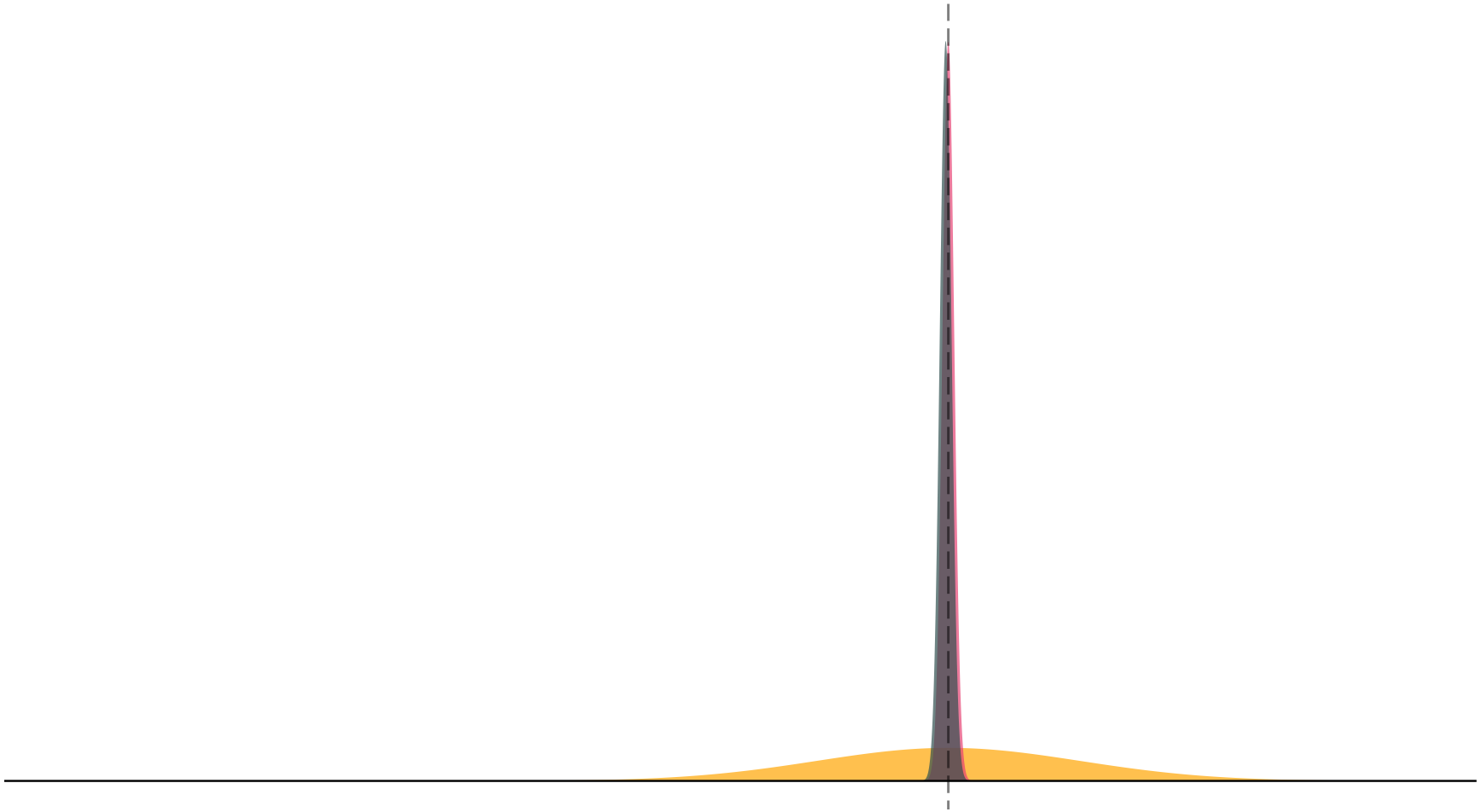
$n = 100$



Consistency

Distributions of X_1, $\overline{X}$, and $\widetilde{X}$

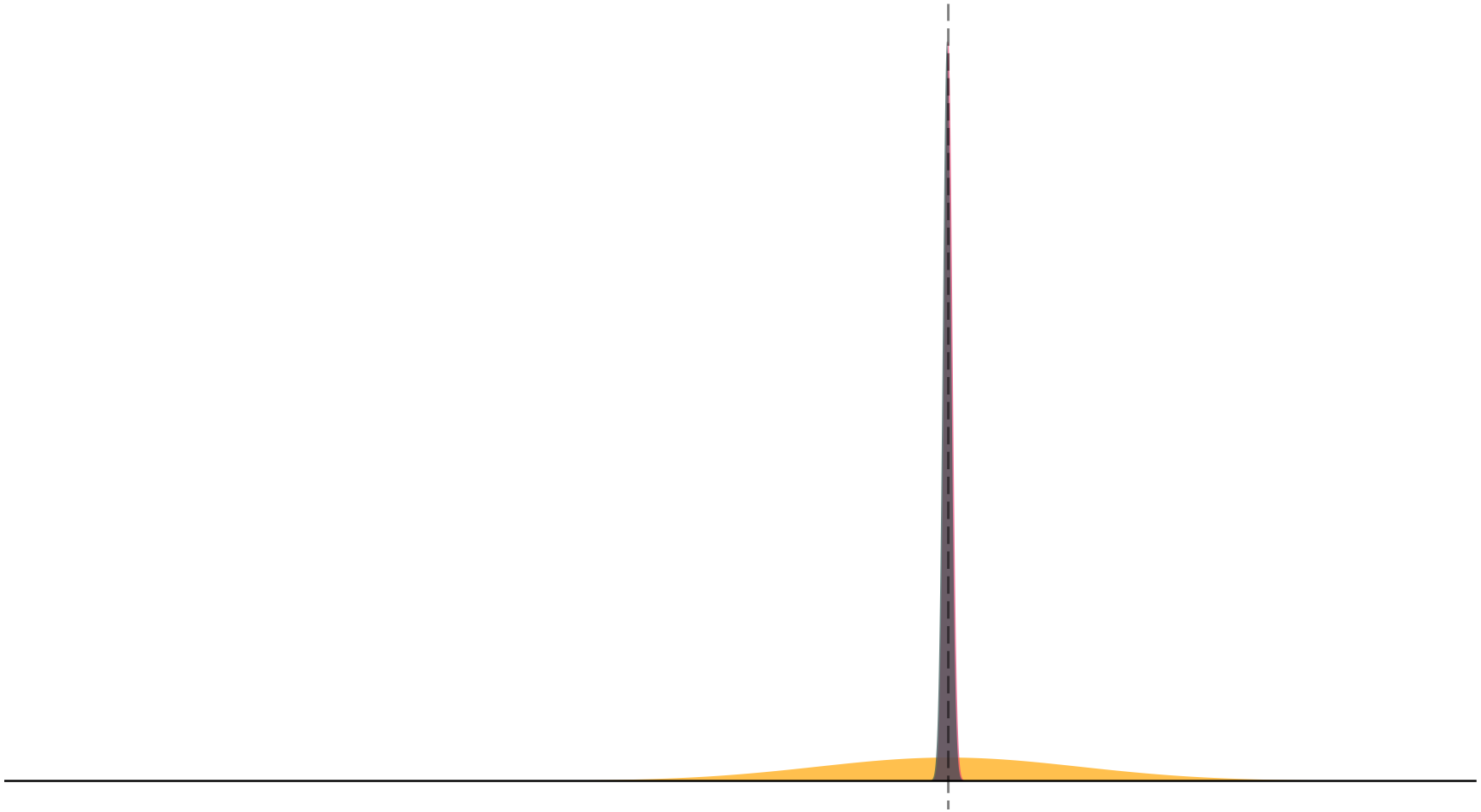
$n = 500$



Consistency

Distributions of X_1, $\overline{X}$, and $\widetilde{X}$

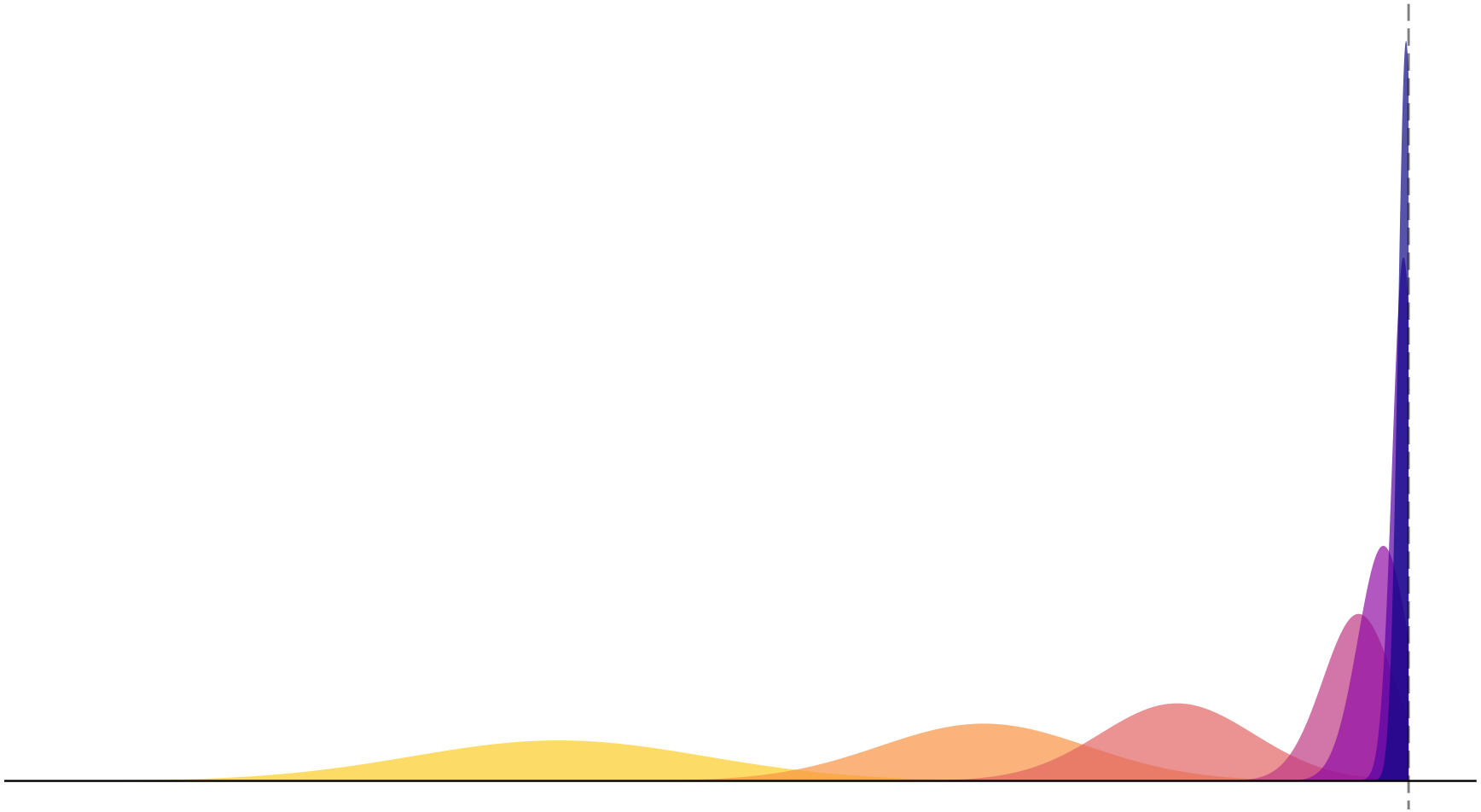
$n = 1000$



Consistency

The distributions of $\widetilde{X}$

For n in $\{2, 5, 10, 50, 100, 500, 1000\}$



Consistency

The takeaway?

Consistency

Why consistency (asymptotics)?

1. We cannot always find an unbiased estimator. In these situations, we generally (at least) want consistency.
2. Expected values can be hard/undefined. Probability limits are less constrained, *e.g.*,

$$\boldsymbol{E}\left[g(X)h(Y)\right] \text{ vs. } \operatorname{plim}\left(g(X)h(Y)\right)$$

3. Asymptotics help us move away from assuming the distribution of $\boldsymbol{u}_i$.

OLS in asymptopia

OLS in asymptopia

OLS has two very nice asymptotic properties:

1. Consistency
2. Asymptotic Normality

OLS in asymptopia

Proof of consistency

First, recall our previous derivation of of $\hat{\beta}_1$,

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_i (x_i - \overline{x}) u_i}{\sum_i (x_i - \overline{x})^2}$$

OLS in asymptopia

Proof of consistency

We actually want to know the probability limit of $\hat{\beta}_1$, so

OLS in asymptopia

Proof of consistency

So we have

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 + \frac{\operatorname{Cov}(x,\, u)}{\operatorname{Var}(x)}$$

OLS in asymptopia

Asymptotic normality

Up to this point, we made a very specific assumption about the distribution of u_i—the u_i came from a normal distribution.

We can relax this assumption—allowing the u_i to come from any distribution (still assume exogeneity, independence, and homoskedasticity).

We will focus on the **asymptotic distribution** of our estimators (how they are distributed as n gets large), rather than their finite-sample distribution.

OLS in asymptopia

Recap

With a more limited set of assumptions, OLS is **consistent** and is **asymptotically normally distributed**.

Current assumptions

1. Our data were **randomly sampled** from the population.
2. y_i is a **linear function** of its parameters and disturbance.
3. There is **no perfect collinearity** in our data.
4. The u_i have conditional mean of zero (**exogeneity**), $\boldsymbol{E}\left[u_i | X_i\right] = 0$.
5. The u_i are **homoskedastic** with **zero correlation** between u_i and u_j.

Omitted-variable bias, redux

Omitted-variable bias, redux

Inconsistency?

Imagine we have a population whose true model is

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i \tag{2}$$

Omitted-variable bias, redux

Inconsistency?

Imagine we have a population whose true model is

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i \tag{2}$$

Recall$_2$: We defined the **bias** of an estimator W for parameter θ

Omitted-variable bias, redux

Inconsistency?

Imagine we have a population whose true model is

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i \tag{2}$$

We know that omitted-variable bias causes biased estimates.

Question: Do *omitted variables* also cause inconsistent estimates?

Omitted-variable bias, redux

Inconsistency?

Imagine we have a population whose true model is

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i \tag{2}$$

but we instead specify the model as

$$y_i = \beta_0 + \beta_1 x_{1i} + w_i \tag{3}$$

where $w_i = \beta_2 x_{2i} + u_i$.

Omitted-variable bias, redux

Inconsistency?

Truth: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$ **Specified:** $y_i = \beta_0 + \beta_1 x_{1i} + w_i$

We already showed $\text{plim}\, \hat{\beta}_1 = \beta_1 + \dfrac{\text{Cov}\left(x_1,\, w\right)}{\text{Var}\left(x_1\right)}$

where w is the disturbance.

Omitted-variable bias, redux

Inconsistency?

$$\text{plim}\,\hat{\beta}_1 = \beta_1 + \frac{\text{Cov}\left(x_1,\, \beta_2 x_2\right) + \text{Cov}\left(x_1,\, u\right)}{\text{Var}\left(x_1\right)}$$

Now we use the fact that $\text{Cov}\left(X,\, cY\right) = c\,\text{Cov}\left(X,\, Y\right)$ for a constant c.

Omitted-variable bias, redux

Inconsistency?

Thus, we find that

$$\text{plim}\,\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\text{Cov}\left(x_1,\, x_2\right)}{\text{Var}\left(x_1\right)}$$

In other words, an omitted variable will cause OLS to be inconsistent if **both** of the following statements are true:

1. The omitted variable **affects our outcome**, *i.e.*, $\beta_2 \neq 0$.
2. The omitted variable **correlates with included explanatory variables**, *i.e.*, $\text{Cov}\left(x_1,\, x_2\right) \neq 0$.

If both of these statements are true, then the OLS estimate $\hat{\beta}_1$ will not converge to β_1, even as n approaches ∞.

Omitted-variable bias, redux

Signing the bias

Sometimes we're stuck with omitted variable bias.[1]

$$\text{plim}\,\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\text{Cov}\left(x_1,\, x_2\right)}{\text{Var}\left(x_1\right)}$$

When this happens, we can often at least know the direction of the inconsistency.

1. You will often hear the term "omitted-variable bias" when we're actually talking about inconsistency (rather than bias).

Omitted-variable bias, redux

Signing the bias

Begin with

$$\text{plim}\,\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\text{Cov}\left(x_1,\, x_2\right)}{\text{Var}\left(x_1\right)}$$

We know $\text{Var}\left(x_1\right) > 0$. Suppose $\beta_2 > 0$ and $\text{Cov}\left(x_1,\, x_2\right) > 0$. Then

Omitted-variable bias, redux

Signing the bias

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Omitted-variable bias, redux

Signing the bias

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We know $\text{Var}\left(x_1\right) > 0$. Suppose $\beta_2 > 0$ and $\text{Cov}\left(x_1,\, x_2\right) < 0$. Then

$$\text{plim}\,\hat{\beta}_1 = \beta_1 + (+)\frac{(-)}{(+)} \implies \text{plim}\,\hat{\beta}_1 < \beta_1$$

$\therefore$ In this case, OLS is **biased downward** (estimates are too small).

	$\text{Cov}(x_1,\, x_2) > 0$	$\text{Cov}(x_1,\, x_2) < 0$
$\beta_2 > 0$	Upward	Downward
$\beta_2 < 0$	Downward	

Omitted-variable bias, redux

Signing the bias

Begin with

$$\text{plim}\,\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\text{Cov}\left(x_1,\, x_2\right)}{\text{Var}\left(x_1\right)}$$

We know $\text{Var}\left(x_1\right) > 0$. Suppose $\beta_2 < 0$ and $\text{Cov}\left(x_1,\, x_2\right) < 0$. Then

$$\text{plim}\,\hat{\beta}_1 = \beta_1 + (-)\frac{(-)}{(+)} \implies \text{plim}\,\hat{\beta}_1 > \beta_1$$

$\therefore$ In this case, OLS is **biased upward** (estimates are too large).

	$\text{Cov}(x_1,\, x_2) > 0$	$\text{Cov}(x_1,\, x_2) < 0$
$\beta_2 > 0$	Upward	Downward
$\beta_2 < 0$	Downward	Upward

Omitted-variable bias, redux

Signing the bias

Thus, in cases where we have a sense of

1. the sign of $\text{Cov}\left(x_1,\, x_2\right)$
2. the sign of β_2

we know in which direction inconsistency pushes our estimates.

Direction of bias

	$\text{Cov}(x_1,\, x_2) > 0$	$\text{Cov}(x_1,\, x_2) < 0$
$\beta_2 > 0$	Upward	Downward
$\beta_2 < 0$	Downward	Upward

Measurement error

Measurement error in our explanatory variables presents another case in which OLS is inconsistent.

Consider the population model: $y_i = \beta_0 + \beta_1 z_i + u_i$

- We want to observe z_i but cannot.
- Instead, we *measure* the variable x_i, which is z_i plus some error (noise):

$$x_i = z_i + \omega_i$$

- Assume $\boldsymbol{E}\left[\omega_i\right] = 0$, $\mathrm{Var}\left(\omega_i\right) = \sigma_\omega^2$, and ω is independent of z and u.

Measurement error

Proof

$$y_i = \beta_0 + \beta_1 z_i + u_i$$

Measurement error

Proof

The covariance of our noisy variable x and the disturbance ε.

$$\mathrm{Cov}\left(x,\, \varepsilon\right)$$

Measurement error

Proof

Now for the denominator, $\mathbf{Var}\left(x\right)$.

$\mathbf{Var}\left(x\right)$

Measurement error

Proof

Putting the numerator and denominator back together,

$$
\begin{aligned}
\operatorname{plim} \hat{\beta}_1 &= \beta_1 + \frac{\operatorname{Cov}(x, \varepsilon)}{\operatorname{Var}(x)} \\
&= \beta_1 + \frac{-\beta_1 \sigma_\omega^2}{\sigma_z^2 + \sigma_\omega^2} \\
&= \beta_1 - \beta_1 \frac{\sigma_\omega^2}{\sigma_z^2 + \sigma_\omega^2} \\
&= \beta_1 \frac{\sigma_z^2 + \sigma_\omega^2}{\sigma_z^2 + \sigma_\omega^2} - \beta_1 \frac{\sigma_\omega^2}{\sigma_z^2 + \sigma_\omega^2} \\
&= \beta_1 \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\omega^2}
\end{aligned}
$$

Measurement error

Summary

$$\therefore \operatorname{plim} \hat{\beta}_1 = \beta_1 \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\omega^2}.$$

What does this equation tell us?

Measurement error

Summary

What about **measurement in the outcome variable**?

It doesn't really matter—it just increases our standard errors.

Measurement error

It's everywhere

General cases

1. We cannot perfectly observe a variable.
2. We use one variable as a *proxy* for another.

Specific examples

- GDP
- Population
- Crime/police statistics
- Air quality
- Health data
- Ability

Conclusions

Wrapping up

Asymptotic properties highlight some interesting features:

- **Consistency** describes estimators' behavior as n approaches ∞.
- **Omitted-variable bias** causes OLS to be inconsistent.

$\Longrightarrow$ Even if with *infinite data*, you can get the wrong answer.